

Linear Algebra I — Week 3

Span, Linear Independence & Setting Up Linear Systems

Name: _____ Date: _____ All work **by hand** — no calculator, no software.

Key Facts

Linear combination and span. A linear combination of $v_1, \dots, v_k \in V$ is $a_1v_1 + \dots + a_kv_k$ with $a_i \in \mathbb{F}$. $\text{span}\{v_1, \dots, v_k\}$ is the set of all of them; $\text{span} \emptyset = \{\mathbf{0}\}$. It is a subspace, and the smallest subspace containing $v_1, \dots, v_k$.

Setting up the system. To test $w \in \text{span}\{v_1, \dots, v_k\}$ in $\mathbb{F}^n$, solve $a_1v_1 + \dots + a_kv_k = w$. Coordinatewise this is n equations in k unknowns, with augmented matrix whose columns are $v_1, \dots, v_k \mid w$. Solution exists $\iff w \in \text{span}$.

Linear independence. $v_1, \dots, v_k$ are *independent* if $a_1v_1 + \dots + a_kv_k = \mathbf{0}$ forces $a_1 = \dots = a_k = 0$; otherwise *dependent*. Test: solve the homogeneous system; only $\mathbf{0} \Rightarrow$ independent, a nonzero solution $\Rightarrow$ dependent (and it gives a relation).

Useful facts. A set containing $\mathbf{0}$ is dependent. One vector is independent iff it is nonzero. Two vectors are independent iff neither is a scalar multiple of the other. $v_1, \dots, v_k$ dependent $\iff$ some $v_j \in \text{span}$ of the others. If $w \in \text{span}\{v_1, \dots, v_k\}$, adding w does not enlarge the span. Independence and span depend on the field: $\{(1, i), (i, -1)\}$ is dependent over $\mathbb{C}$ but independent over $\mathbb{R}$.

Counting. More homogeneous unknowns than equations forces a nonzero solution; so more than n vectors in $\mathbb{F}^n$ are dependent.

Worked Examples

Example 1 · Membership in a span

Is $(4, 7) \in \text{span}\{(1, 2), (3, 5)\}$ in $\mathbb{R}^2$?

Solution.

1. Solve $a(1, 2) + b(3, 5) = (4, 7)$: $a + 3b = 4$, $2a + 5b = 7$.
2. From the first, $a = 4 - 3b$. Substitute: $8 - 6b + 5b = 7$, so $b = 1$, $a = 1$.
3. Check: $(1, 2) + (3, 5) = (4, 7)$.

Answer: Yes: $(4, 7) = 1(1, 2) + 1(3, 5)$.

Example 2 · Membership (positive)

Is $(1, 2, 3) \in \text{span}\{(1, 0, 1), (0, 1, 1)\}$?

Solution.

1. $a(1, 0, 1) + b(0, 1, 1) = (a, b, a + b)$.
2. Match: $a = 1$, $b = 2$; third coordinate $a + b = 3$ ✓.

Answer: Yes: $(1, 2, 3) = 1(1, 0, 1) + 2(0, 1, 1)$.

Example 3 · Membership (negative)

Is $(1, 2, 4) \in \text{span}\{(1, 0, 1), (0, 1, 1)\}$?

Solution.

1. As before the combination is $(a, b, a + b)$.
2. Match: $a = 1$, $b = 2$, but then $a + b = 3 \neq 4$.

Answer: No.

Example 4 · Dependence, finding a relation

Test $\{(1, 2, 1), (2, 1, 0), (1, -1, -1)\}$ for independence.

Solution.

1. Solve $a(1, 2, 1) + b(2, 1, 0) + c(1, -1, -1) = \mathbf{0}$: $a + 2b + c = 0$, $2a + b - c = 0$, $a - c = 0$.
2. Third: $c = a$. First: $2a + 2b = 0$, so $b = -a$. Second: $2a - a - a = 0$ ✓ (automatically).
3. So $(a, b, c) = (1, -1, 1)$ is a nontrivial solution. Check: $v_1 - v_2 + v_3 = (1 - 2 + 1, 2 - 1 - 1, 1 - 0 - 1) = \mathbf{0}$.

Answer: Linearly dependent: $v_1 - v_2 + v_3 = \mathbf{0}$.

Example 5 · Independence in P_2

Show $\{1 + x, x + x^2, 1 + x^2\}$ is linearly independent in $P_2(\mathbb{R})$.

Solution.

1. Suppose $a(1 + x) + b(x + x^2) + c(1 + x^2) = \mathbf{0}$. Compare coefficients.
2. Constant: $a + c = 0$. x : $a + b = 0$. x^2 : $b + c = 0$.
3. So $a = -c$ and $b = -a = c$; then $b + c = 2c = 0$, so $c = 0$, hence $a = b = 0$.

Answer: Linearly independent.

Example 6 · The field matters

Is $\{(1, i), (i, -1)\}$ linearly independent in $\mathbb{C}^2$ over $\mathbb{C}$? Over $\mathbb{R}$?

Solution.

1. Note $i \cdot (1, i) = (i, i^2) = (i, -1)$.
2. Over $\mathbb{C}$: $i v_1 - v_2 = \mathbf{0}$, a nontrivial relation, so dependent.
3. Over $\mathbb{R}$: $a(1, i) + b(i, -1) = \mathbf{0}$ with real a, b gives first coordinate $a + bi = 0$, so $a = b = 0$: independent.

Answer: Dependent over $\mathbb{C}$; independent over $\mathbb{R}$.

Example 7 · Proof: dependence means one vector is redundant

Prove: if $v_1, \dots, v_k$ are linearly dependent, then some v_j is a linear combination of the others.

Solution.

1. Take a nontrivial relation $a_1 v_1 + \dots + a_k v_k = \mathbf{0}$ and pick j with $a_j \neq 0$.
2. Isolate: $a_j v_j = -\sum_{i \neq j} a_i v_i$.
3. Divide by a_j (allowed because $a_j \neq 0$ in a field): $v_j = -\sum_{i \neq j} \frac{a_i}{a_j} v_i$.

Answer: $v_j \in \text{span}\{v_i : i \neq j\}$.

Example 8 · Describing a span by an equation

Find the condition on (a, b, c) for $(a, b, c) \in \text{span}\{(1, 1, 0), (0, 1, 1)\}$.

Solution.

1. $x(1, 1, 0) + y(0, 1, 1) = (x, x + y, y)$.
2. Matching: $x = a$, $y = c$, and $b = x + y = a + c$.

Answer: $a - b + c = 0$

Practice Problems

Part A — Linear Combinations and Span

1. [Foundational]
Write $(5, -1)$ as a linear combination of $(1, 1)$ and $(1, -1)$ in $\mathbb{R}^2$.

Step 1. Set $a(1, 1) + b(1, -1) = (5, -1)$: $a + b = 5$ and $a - b = -1$.

Step 2. Add the equations: $2a = 4$, so $a = 2$. Then $b = 5 - 2 = 3$.

Step 3. Check: $2(1, 1) + 3(1, -1) = (2 + 3, 2 - 3) = (5, -1)$.

Answer: $(5, -1) = 2(1, 1) + 3(1, -1)$

2. [Foundational]
Decide whether $(2, 3, 4) \in \text{span}\{(1, 1, 1), (0, 1, 2)\}$ in $\mathbb{R}^3$. If so, give the coefficients.

Step 1. $a(1, 1, 1) + b(0, 1, 2) = (a, a + b, a + 2b)$. Set equal to $(2, 3, 4)$.

Step 2. First coordinate: $a = 2$. Second: $2 + b = 3$, so $b = 1$.

Step 3. Third coordinate check: $a + 2b = 2 + 2 = 4$ ✓.

Answer: Yes: $(2, 3, 4) = 2(1, 1, 1) + 1(0, 1, 2)$.

3. [Foundational]
Decide whether $(1, 0, 0) \in \text{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$.

Step 1. $x(1, 1, 0) + y(0, 1, 1) = (x, x + y, y)$. Set equal to $(1, 0, 0)$.

Step 2. First coordinate: $x = 1$. Third: $y = 0$.

Step 3. Second coordinate: $x + y = 1 \neq 0$. Contradiction.

Answer: No, $(1, 0, 0) \notin \text{span}\{(1, 1, 0), (0, 1, 1)\}$.

4. [Intermediate]
Find k such that $(1, 2, k) \in \text{span}\{(1, 1, 0), (1, -1, 2)\}$.

Step 1. $a(1, 1, 0) + b(1, -1, 2) = (a + b, a - b, 2b)$. Set equal to $(1, 2, k)$.

Step 2. $a + b = 1$ and $a - b = 2$: adding gives $2a = 3$, so $a = \frac{3}{2}$; then $b = -\frac{1}{2}$.

Step 3. Third coordinate: $k = 2b = -1$.

Step 4. Check: $\frac{3}{2}(1, 1, 0) - \frac{1}{2}(1, -1, 2) = (1, 2, -1)$ ✓.

Answer: $k = -1$

5. [Intermediate]
In $\mathbb{C}^2$ over $\mathbb{C}$, write $(1 + i, 2)$ as a linear combination of $(1, i)$ and $(1, -i)$.

Step 1. Solve $a(1, i) + b(1, -i) = (1 + i, 2)$: $a + b = 1 + i$ and $ia - ib = 2$.

Step 2. Divide the second equation by i : $a - b = \frac{2}{i} = -2i$.

Step 3. Add: $2a = 1 + i - 2i = 1 - i$, so $a = \frac{1 - i}{2}$. Subtract: $2b = 1 + i + 2i = 1 + 3i$, so $b = \frac{1 + 3i}{2}$.

Step 4. Check: $a + b = \frac{2 + 2i}{2} = 1 + i$ ✓ and $a - b = \frac{-4i}{2} = -2i$ ✓.

Answer: $a = \frac{1 - i}{2}$, $b = \frac{1 + 3i}{2}$

6.

[Foundational]

Show that $\text{span}\{(1, 2), (2, 4)\} \subseteq \mathbb{R}^2$ is a line, and give its equation.

Step 1. A general element is $a(1, 2) + b(2, 4) = (a + 2b, 2a + 4b) = (a + 2b)(1, 2)$.

Step 2. Put $t = a + 2b$; as a, b vary, t takes every real value.

Step 3. So the span is $\{(t, 2t) : t \in \mathbb{R}\}$, i.e. all points with $y = 2x$.

Answer: The line $y = 2x$ through the origin.

7.

[Intermediate]

Describe $\text{span}\{(1, 0, 2), (0, 1, -1)\} \subseteq \mathbb{R}^3$ by a single linear equation in x, y, z .

Step 1. $a(1, 0, 2) + b(0, 1, -1) = (a, b, 2a - b)$.

Step 2. So $x = a, y = b, z = 2a - b = 2x - y$.

Step 3. Rearrange: $2x - y - z = 0$. Conversely every $(x, y, 2x - y)$ is obtained with $a = x, b = y$.

Answer: $\{(x, y, z) : 2x - y - z = 0\}$

8.

[Challenge]

Prove that for any vectors $v_1, \dots, v_k$ in a vector space V , the set $\text{span}\{v_1, \dots, v_k\}$ is a subspace of V .

Step 1. Zero vector: $\mathbf{0} = 0v_1 + \dots + 0v_k$, so $\mathbf{0} \in \text{span}$.

Step 2. Addition: $\sum a_i v_i + \sum b_i v_i = \sum (a_i + b_i) v_i$, again a linear combination.

Step 3. Scalars: $\lambda \sum a_i v_i = \sum (\lambda a_i) v_i$, again a linear combination.

Step 4. All three subspace conditions hold.

Answer: $\text{span}\{v_1, \dots, v_k\}$ is a subspace of V .

Part B — Linear Independence and Dependence

9.

[Foundational]

Is $\{(1, 2), (3, 6)\}$ linearly independent in $\mathbb{R}^2$?

Step 1. Observe $(3, 6) = 3(1, 2)$.

Step 2. Hence $3(1, 2) - 1(3, 6) = \mathbf{0}$ is a nontrivial relation.

Answer: Linearly dependent.

10.

[Foundational]

Is $\{(1, 2), (2, 3)\}$ linearly independent in $\mathbb{R}^2$?

Step 1. Solve $a(1, 2) + b(2, 3) = \mathbf{0}$: $a + 2b = 0$ and $2a + 3b = 0$.

Step 2. From the first, $a = -2b$. Substitute: $-4b + 3b = -b = 0$, so $b = 0$ and $a = 0$.

Step 3. Only the trivial solution.

Answer: Linearly independent.

11.

[Intermediate]

 Is $\{(1, 0, 1), (1, 1, 0), (0, 1, 1)\}$ linearly independent in $\mathbb{R}^3$?

Step 1. Solve $a(1, 0, 1) + b(1, 1, 0) + c(0, 1, 1) = \mathbf{0}$: $a + b = 0$, $b + c = 0$, $a + c = 0$.

Step 2. So $a = -b$ and $c = -b$. Then $a + c = -2b = 0$, giving $b = 0$.

Step 3. Hence $a = b = c = 0$.

Answer: Linearly independent.

12.

[Intermediate]

 Is $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ linearly independent in $\mathbb{R}^3$? If not, give a nontrivial relation.

Step 1. Solve $av_1 + bv_2 + cv_3 = \mathbf{0}$: $a + 4b + 7c = 0$, $2a + 5b + 8c = 0$, $3a + 6b + 9c = 0$.

Step 2. Eq. 2 – Eq. 1: $a + b + c = 0$. Eq. 3 – Eq. 2: $a + b + c = 0$ (same equation). So the system has a free variable.

Step 3. Eq. 1 – $(a + b + c)$: $3b + 6c = 0$, so $b = -2c$; then $a = -b - c = c$. Take $c = 1$: $(a, b, c) = (1, -2, 1)$.

Step 4. Check: $v_1 - 2v_2 + v_3 = (1 - 8 + 7, 2 - 10 + 8, 3 - 12 + 9) = (0, 0, 0) \checkmark$.

Answer: Linearly dependent: $v_1 - 2v_2 + v_3 = \mathbf{0}$.

13.

[Challenge]

 Find all real k for which $\{(1, 1, 1), (1, 2, 3), (1, 3, k)\}$ is linearly dependent.

Step 1. Solve $a(1, 1, 1) + b(1, 2, 3) + c(1, 3, k) = \mathbf{0}$: $a + b + c = 0$, $a + 2b + 3c = 0$, $a + 3b + kc = 0$.

Step 2. Eq. 2 – Eq. 1: $b + 2c = 0$. Eq. 3 – Eq. 2: $b + (k - 3)c = 0$.

Step 3. Subtract these two: $(k - 5)c = 0$.

Step 4. If $k \neq 5$: $c = 0$, then $b = 0$, $a = 0$ (independent). If $k = 5$: c is free, e.g. $c = 1$, $b = -2$, $a = 1$ (dependent).

Answer: Dependent exactly when $k = 5$.

14.

[Foundational]

 In $P_2(\mathbb{R})$, is $\{1 + x, 1 - x, 2\}$ linearly independent?

Step 1. Add the first two: $(1 + x) + (1 - x) = 2$.

Step 2. So $(1 + x) + (1 - x) - 1 \cdot 2 = \mathbf{0}$ is a nontrivial relation.

Answer: Linearly dependent.

15.

[Intermediate]

 In $\mathbb{C}^2$, consider $\{(1, i), (1 + i, i - 1)\}$. (a) Is it linearly independent over $\mathbb{C}$? (b) Is it linearly independent over $\mathbb{R}$?

Step 1. (a) Compute $(1 + i)(1, i) = (1 + i, i + i^2) = (1 + i, i - 1)$, which is the second vector.

Step 2. So $(1 + i)v_1 - v_2 = \mathbf{0}$: dependent over $\mathbb{C}$.

Step 3. (b) Take real a, b with $a(1, i) + b(1 + i, i - 1) = \mathbf{0}$. First coordinate: $a + b + bi = 0$.

Step 4. Imaginary part gives $b = 0$; then real part gives $a = 0$. Only the trivial solution over $\mathbb{R}$.

Answer: (a) dependent over $\mathbb{C}$; (b) independent over $\mathbb{R}$.

16.

[Challenge]

Let u, v, w be linearly independent in a real vector space. Prove that $u + v$, $v + w$, $u + w$ are linearly independent.

Step 1. Suppose $a(u + v) + b(v + w) + c(u + w) = \mathbf{0}$.

Step 2. Regroup: $(a + c)u + (a + b)v + (b + c)w = \mathbf{0}$.

Step 3. By independence of u, v, w : $a + c = 0$, $a + b = 0$, $b + c = 0$.

Step 4. Adding: $2(a + b + c) = 0$, so $a + b + c = 0$. Then $a = (a + b + c) - (b + c) = 0$, and similarly $b = 0$, $c = 0$.

Answer: Only the trivial relation exists, so they are linearly independent.

Part C — Setting Up Systems, Spanning Sets and Proofs

17.

[Foundational]

Set up and solve the system that decides whether $(1, 2, 3) \in \text{span}\{(1, 0, 1), (2, 1, 0), (0, 1, 1)\}$.

Step 1. Seek a, b, c with $a(1, 0, 1) + b(2, 1, 0) + c(0, 1, 1) = (1, 2, 3)$. Coordinatewise: $a + 2b = 1$, $b + c = 2$, $a + c = 3$.

Step 2. Augmented matrix (columns are the three vectors, last column is the target): $\left(\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 \end{array} \right)$.

Step 3. From Eq. 1: $a = 1 - 2b$. From Eq. 2: $c = 2 - b$. Substitute into Eq. 3: $3 - 3b = 3$, so $b = 0$, $a = 1$, $c = 2$.

Step 4. Check: $1(1, 0, 1) + 0(2, 1, 0) + 2(0, 1, 1) = (1, 2, 3) \checkmark$.

Answer: Yes: $(1, 2, 3) = 1(1, 0, 1) + 0(2, 1, 0) + 2(0, 1, 1)$.

18.

[Intermediate]

Find the condition on (a, b, c) for $(a, b, c) \in \text{span}\{(1, 2, 0), (0, 1, 1)\}$.

Step 1. $x(1, 2, 0) + y(0, 1, 1) = (x, 2x + y, y)$.

Step 2. Matching (a, b, c) : $x = a$, $y = c$, and $b = 2x + y = 2a + c$.

Step 3. The condition is $2a - b + c = 0$; conversely, if it holds take $x = a$, $y = c$.

Answer: $2a - b + c = 0$

19.

[Intermediate]

Find a finite spanning set for $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$.

Step 1. Solve for x : $x = -y - z$, so every element is $(-y - z, y, z)$.

Step 2. Split: $(-y - z, y, z) = y(-1, 1, 0) + z(-1, 0, 1)$.

Step 3. Hence $W \subseteq \text{span}\{(-1, 1, 0), (-1, 0, 1)\}$; both vectors satisfy $x + y + z = 0$, so equality holds.

Answer: $W = \text{span}\{(-1, 1, 0), (-1, 0, 1)\}$

20.

[Intermediate]

 (a) Show that $\{(1, 1), (1, 2), (2, 1)\}$ spans $\mathbb{R}^2$. (b) Is it linearly independent?

Step 1. (a) For a target (x, y) use only the first two: $a + b = x$, $a + 2b = y$. Subtract: $b = y - x$, then $a = 2x - y$. Every (x, y) is reached.

Step 2. (b) Look for a relation among all three. Try $(2, 1) = a(1, 1) + b(1, 2)$: $a + b = 2$, $a + 2b = 1$ give $b = -1$, $a = 3$.

Step 3. Check: $3(1, 1) - (1, 2) = (2, 1) \checkmark$, so $3(1, 1) - (1, 2) - (2, 1) = \mathbf{0}$.

Answer: (a) Yes it spans. (b) No, it is linearly dependent.

21.

[Intermediate]

 Show that $\{1, 1 + x, 1 + x + x^2\}$ spans $P_2(\mathbb{R})$.

Step 1. Let $a + bx + cx^2$ be arbitrary. Seek α, β, γ with $a + bx + cx^2 = \alpha \cdot 1 + \beta(1 + x) + \gamma(1 + x + x^2)$.

Step 2. Compare coefficients: x^2 : $\gamma = c$. x : $\beta + \gamma = b$, so $\beta = b - c$. Constant: $\alpha + \beta + \gamma = a$, so $\alpha = a - b$.

Step 3. Such α, β, γ exist for every a, b, c .

Answer: $a + bx + cx^2 = (a - b) \cdot 1 + (b - c)(1 + x) + c(1 + x + x^2)$; spans.

22.

[Challenge]

 Show that $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\}$ spans $M_{2 \times 2}(\mathbb{R})$ and is linearly independent.

Step 1. Let $\begin{pmatrix} p & q \\ r & s \end{pmatrix} = \alpha I + \beta \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \delta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Step 2. Entries give: $p = \alpha + \delta$, $s = \alpha - \delta$, $q = \beta + \gamma$, $r = \beta - \gamma$.

Step 3. Solve: $\alpha = \frac{p+s}{2}$, $\delta = \frac{p-s}{2}$, $\beta = \frac{q+r}{2}$, $\gamma = \frac{q-r}{2}$. So every matrix is reached: spanning.

Step 4. Independence: if the matrix is $\mathbf{0}$ then $p = q = r = s = 0$, so the formulas give $\alpha = \beta = \gamma = \delta = 0$.

Answer: The set spans $M_{2 \times 2}(\mathbb{R})$ and is linearly independent.

23.

[Challenge]

 Prove: if $w \in \text{span}\{v_1, \dots, v_k\}$ then $\text{span}\{v_1, \dots, v_k, w\} = \text{span}\{v_1, \dots, v_k\}$.

Step 1. ($\supseteq$) Any combination of $v_1, \dots, v_k$ is a combination of $v_1, \dots, v_k, w$ with coefficient 0 on w .

Step 2. Write $w = \sum c_i v_i$. ($\subseteq$) Take $\sum a_i v_i + bw$.

Step 3. Substitute: $\sum a_i v_i + b \sum c_i v_i = \sum (a_i + bc_i) v_i$, a combination of $v_1, \dots, v_k$.

Step 4. Both inclusions hold.

Answer: The two spans are equal.

24.

[Challenge]

 Let $v_1, \dots, v_k$ be linearly independent and $v_{k+1} \notin \text{span}\{v_1, \dots, v_k\}$. Prove that $v_1, \dots, v_{k+1}$ are linearly independent.

Step 1. Suppose $a_1 v_1 + \dots + a_k v_k + a_{k+1} v_{k+1} = \mathbf{0}$.

Step 2. If $a_{k+1} \neq 0$: $v_{k+1} = -\sum_{i \leq k} \frac{a_i}{a_{k+1}} v_i \in \text{span}\{v_1, \dots, v_k\}$, a contradiction.

Step 3. So $a_{k+1} = 0$, leaving $a_1 v_1 + \dots + a_k v_k = \mathbf{0}$.

Step 4. By independence of $v_1, \dots, v_k$: $a_1 = \dots = a_k = 0$.

Answer: All coefficients vanish, so $v_1, \dots, v_{k+1}$ are linearly independent.